MATH808Q - Abelian varieties

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1 Introduction to Abelian varieties

Motivating example: $V \cong \mathbb{C}^g$, $\Lambda \leq V$ is a lattice(discrete cocompact subgroup), $V \otimes_{\mathbb{Z}} \mathbb{R} = V$, $G = \frac{V}{\Lambda}$ is a compact complex abelian Lie group, a complex torus since $\frac{V}{\Lambda} \cong \left(S^1\right)^{2g}$

Theorem 1.1. Any compact connected(connected is necessary, otherwise finite groups would all be abelian) complex Lie group X is abelian

Proof. $n = \dim_{\mathbb{C}} X$. Let $e \in U \subseteq \mathbb{C}^n$ be an open subset of X, $\varphi_g : X \to X, x \mapsto gxg^{-1}$ induces $\varphi : X \to \mathrm{Aut}(X)$, for a fixed $g \in X$, □

Proof. Consider $\varphi : X \times X \to X, (x, y) \mapsto xyx^{-1}y^{-1}$, open neighborhood V_g of g and open neighborhood W_g of e such that $\varphi\left(V_g, W_g\right) \subseteq U$ and $\varphi(g, e) = e$. Since X is compact, $X = \cup_{i=1}^{r} V_{g_i}$, let $W = \cap W_{g_i}$, then $\varphi(X, W) \subseteq U$, is homomorphic, hence $\varphi(X, W) = e$ □

Every such group would be like $X = \frac{V}{\Lambda}$, $V = \mathrm{Lie}\, X = \tilde{X}$, and $\Lambda = \pi_1(X, e) = H_1(X, \mathbb{Z})$. This is a complex tori, consider $X \xrightarrow{n} X$, let $X[n] \cong \left(\frac{\mathbb{Z}}{n}\mathbb{Z}\right)^{2g}$ denote the kernel which consists of n-torsion elements, this is an isogeny of degree n^{2g}

How to classify all such X, $\varphi X = \frac{V}{\Lambda} \to X' = \frac{V'}{\Lambda'} \Leftrightarrow \tilde{\varphi} : V \to V'$ and $\tilde{\varphi}(V) \subseteq V'$, need to consider something like $\mathrm{GL}_g \frac{\mathbb{C}}{\mathrm{GL}_g}(\mathbb{C}) \cap \mathrm{GL}_{2g}(\mathbb{Z})$

Tate module $T_p A \cong \left(\mathbb{Z}_p\right)^{2g}$

Definition 1.2. F is a perfect field with $F = \overline{F}$, an abelian variety A is a group variety that is proper(irreducible, reduced, connected)

Lemma 1.3 (Rigidity). Suppose X is proper, $f : X \times Y \to Z$ is a morphism such that $f(X \times y_0) = z_0 \in Z$, then $f(X \times y)$ is a point in Z for any $y \in Y$

Proof. Just need to show that there exists a unique map $g : Y \to Z$ such that $f = gp_2$, and it suffices to show that $f = gp_2$ on a nonempty open set of $X \times Y$. Since $f(X \times y_0) = z_0$, there exists nonempty closed $W \subseteq Z$ such that $f^{-1}(W)$ is disjoint from $X \times y_0$, and since X is proper, $X \times Y \to Y$ is closed, so $Y' = p_2(f^{-1}(W))$ is closed, and $V = Y - Y'$ is a nonempty open set in Y containing y_0, for any $y \in V$, $f(X \times y) \subseteq U = Z - W$ open, but $X \times y \cong X$ is proper, hence $f(X \times y)$ is a point, denote as $g(y)$, then $f = gp_2$ on $X \times V \subseteq X \times Y$ □

Remark 1.4. Note that the lemma is false if X is not proper, for example $\mathbb{A}^1 \times \mathbb{A}^1 \to \mathbb{A}^1, (x, y) \mapsto xy$

Theorem 1.5 (Chevalley). There is an exact sequence(in general non split)

$$1 \to H \to G \to A \to 1$$

A is an abelian variety, H is affine

Theorem 1.6. Any abelian variety A is commutative(hence the term "abelian")

Proof. Consider $f : A \times A \to A, (x, y) \mapsto xyx^{-1}y^{-1}$, then $f(A, e) = e$, by Lemma 1.3, $f(A, y) = f(e, y) = e$, thus A is commutative □

Corollary 1.7. A is an abelian variety, G is a group variety, $f : A \to G$ is a morphism with $f(e_A) = e_G$, then f is a group homomorphism

Proof. Consider $\varphi : A \times A \to G$, $(x, y) \mapsto f(x)f(y)f(xy)^{-1}$, then $\varphi(A, e_A) = e_G$, by Lemma 1.3, $\varphi(x, y) = \varphi(e_A, y) = e_G$, hence f is a homomorphism □

Corollary 1.8. A is an abelian variety, X, Y are proper varieties pointed at x_0, y_0, then there is a bijection

$$\mathrm{Hom}(X, A) \times \mathrm{Hom}(Y, A) \to \mathrm{Hom}(X \times Y, A), \quad (f, g) \mapsto f + g$$

Proof. This is injective since $f(x) = h(x, y_0)$, $g(y) = h(x_0, y)$. This is also surjective by define f, g the same way and consider $k(x, y) = h(x, y) - f(x) - g(y)$ □

Definition 1.9. S-group scheme. G is commutative if m is a symmetric

Exercise 1.10. $G \to S$ is separable iff $e : S \to G$ is a closed immersion

Definition 1.11. $f : G \to H$ is a S-group scheme homomorphism if it is compatible with $m, i, e, \ker f$ as the pullback of $G \xrightarrow{f} H$ and $S \xrightarrow{e_H} H$ is again a S-group scheme, and $(\ker f)(T) = \ker(f_T)$

An abelian variety doesn't in general have a Hopf algebra structure, but due to Mumford, we can use Heisenberg group, or Mumford Tate group to help us study and understand

A group scheme can be reduced, for example Spec $\frac{\mathbb{F}_{p[t]}}{t^p}$. However we have

Theorem 1.12 (Oort). char $k = 0$, then any group scheme over k is reduced

X is a compact complex Lie group, $g = \dim X$, $H^0(X, \Omega^1) \cong \mathbb{C}^g$, $H_1(X, \mathbb{Z}) \to H^0(X, \Omega^1)^*$ with the image being a lattice Λ, $X \xrightarrow{\cong} H^0 \frac{(X,\Omega^1)^*}{\Lambda} = \frac{V}{\Lambda}$, $V = \operatorname{Lie} X \cong \tilde{X}$ by sending $p \in X$ to $\int_0^p$

If $\varphi : X_1 \to X_2$ is a holomorphic, $\varphi(0)$, $\varphi\left(e_{X_1}\right) = e_{X_2}$, then φ is a homomorphism
Proof.

$$\begin{array}{ccccc} H_1(X_1,\mathbb{Z}) & \to & H^0(X_1,\Omega^1)^* & \to & \frac{V_1}{\Lambda_1} \leftarrow X_1 \\ \downarrow & & \downarrow & & \downarrow \quad \downarrow \\ H_1(X_2,\mathbb{Z}) & \to & H^0(X_2,\Omega^1)^* & \to & \frac{V_2}{\Lambda_2} \leftarrow X_2 \end{array}$$

□

This means any holomorphic map $\varphi : X_1 \to X_2$ is homomorphism + translation

$$\begin{array}{ccc} & \overset{\phi_{\mathbb{C}}}{\nearrow} & \operatorname{Hom}_{\mathbb{C}}(V_1, V_2) \\ \operatorname{Hom}(X_1, X_2) & \hookrightarrow & \operatorname{Hom}_{\mathbb{Z}}(\Lambda_1, \Lambda_2) \\ & \searrow & \operatorname{Hom}\big(H_1(X, \mathbb{Z}_p), H_1(X, \mathbb{Z}_p)\big) \end{array}$$

$\varphi \in \operatorname{Hom}(X_1, X_2)$ determines $\tilde{\varphi} : V_1 \to V_2$, and $\varphi(\Lambda_1) \subseteq \Lambda_2$, thus makes $\phi_{\mathbb{Z}}$ injective, and $\operatorname{Hom}_{\mathbb{Z}}(\Lambda_1, \Lambda_2)$ is of rank $(2 \dim X_1)(2 \dim X_2)$, hence we know $\operatorname{Hom}(X_1, X_2)$ is finite, in fact, $\operatorname{End}(X)$ will be a free $\mathbb{Z}$-algebra of rank $4 \dim X^2$. We also have $\operatorname{im} \phi \subseteq X_2$ is a subtorus, $\ker \phi$ is a subtorus

Definition 1.13. $\varphi : X_1 \to X_2$ is an isogeny if φ is surjective and $\ker \varphi$ is finite, define $\deg \varphi = |\ker \varphi| < \infty$

Composition of isogeny is again an isogeny, deg multiply

$X_1 \sim X_2$ isogeny is an equivalence relation. Suppose $\varphi : X_1 \to X_2$ is an isogeny of deg n, then there exists a unique isogeny $\varphi^* : X_2 \to X_1$

Consider

$$\begin{array}{ccc} X_1 & \overset{\varphi}{\to} & X_2 \\ & {\scriptstyle n} \searrow & \downarrow {\scriptstyle \varphi^*} \\ & & X_1 \end{array}$$

since homo, we have

$$\begin{array}{ccc} X_{\frac{1}{n}1} & \overset{\varphi}{\to} & X_{\frac{1}{n}2} \\ \downarrow & & \downarrow \\ X_1 & \overset{\varphi}{\to} & X_2 \end{array}$$

$(\varphi\varphi^* - n)\varphi = 0$ and φ is surjective

By Chow's theorem, any analytic variety V in $\mathbb{CP}^n$ is an algebraic variety, and thus a group variety(The graph $m : V \times V \to V$ being $V \times V \times V$ being an analytic variety that can be embedded in $\mathbb{CP}^N$), actually an abelian variety(being proper)

Let V be a dimension g complex vector space, $H : V \times V \to \mathbb{C}$ is a Hermitian form, $H = S + iE$, $S = \operatorname{Re} H$, $E = \operatorname{Im} H$, then

$X = \frac{V}{\Lambda}$ is an abelian variety, $E|_{\Lambda\times\Lambda} : \Lambda \times \Lambda \to \mathbb{Z}$ is a bilinear $\mathbb{Z}$ valued map, then we say that E is a Riemann form

Example 1.14. $X = \frac{\mathbb{C}}{\Lambda}$, where $\Lambda = \mathbb{Z}\lambda_1 + \mathbb{Z}\lambda_2$, define E such that $z \wedge w = E(z, w)\lambda_1 \wedge \lambda_2$,

Example 1.15. Suppose $\frac{K}{\mathbb{Q}}$ is a CM field(totally imaginary field), $\frac{K^+}{\mathbb{Q}}$ is the totally real field. $\Phi : K \otimes_{\mathbb{Q}} \mathbb{R} \to \mathbb{C}^g$ is an embedding. Nonzero ideal $I \subseteq \mathcal{O}_K$ are both lattices, say $\Phi(\mathcal{O}_K)$ is Λ, $X = \frac{\mathbb{C}^g}{\Lambda}$ is called a CM abelian variety, $\mathcal{O}_K \hookrightarrow \mathrm{End}(X)$ are multiplications. It is a fact that $K = K^+(\xi)$ with $-\xi^2 \in K^+$ totally positive and $\mathrm{Im}\big(\varphi_j(\xi)\big) > 0$. For $z, w \in \mathbb{C}^g$, define $E(z,w) = \sum_j \varphi\big(g_j\big)\big(\overline{z}_j w_j - z_j \overline{w}_j\big)$, for $\alpha, \beta \in \mathcal{O}_K$, $E(\Phi(\alpha), \Phi(\beta)) = \mathrm{Tr}_{\frac{K}{\mathbb{Q}}}(\xi \tilde{\alpha} \beta)$, here $K \to K, x \mapsto \tilde{x}$ is the unique automorphism fixing K^+

$$\begin{array}{ccc} V \times \mathbb{C} \cong \pi^* L & \to & L \\ \downarrow & & \downarrow \\ V & \overset{\pi}{\to} & X \end{array}$$

$\pi^* L$ is holomorphically trivial because the exponential sequence $0 \to 2\pi i\mathbb{Z} \to \mathcal{O}_V \to \mathcal{O}_V^\times \to 0$

Consider the group action of Λ on V, and corresponding action Λ on $V \times \mathbb{C}$ defined as $(z,t) \cdot \lambda = (z + \lambda, e_\lambda(z)t)$, here $e_\lambda \in \mathbb{O}_V^\times$ satisfying the cocycle condition $e_{\lambda_1+\lambda_2}(z) = e_{\lambda_2(z+\lambda_1)} e_{\lambda_1}(z)$, note that this is commutative

Let F be a number field, with r_1 real embeddings and $2r_2$ complex embeddings, $r_1 + 2r_2 = [F : \mathbb{Q}]$, let w_F be the number of roots of unity, $h_F = |\operatorname{Cl}(\mathcal{O}_F)|$ be the class number of F, r_F is the regulator of F which is the covolume of the lattice $\frac{\mathcal{O}_F^\times}{\text{torsion}} \hookrightarrow \mathbb{R}^{r_1+r_2-1}$. We can the define the zeta function

$$\zeta_F(s) = \sum_{0 \neq \mathfrak{I} \leq \mathcal{O}_F} \frac{1}{|\mathfrak{I}|^s} = \prod_{\mathfrak{p} \text{ prime}} \frac{1}{1 - |\mathfrak{p}|^s}, \quad \mathrm{Re}\, s > 1$$

Theorem 1.16. $\zeta_F(s)$ admits analytic continuation to all of $\mathbb{C}$ with a simple pole at $s = 1$, so $(s-1)\zeta_F(s)$ is analytic, and it satisfies the functional relation $\zeta_F(s) = 2^s \pi^{s-1} \sin\big(\frac{\pi s}{2}\big)\Gamma(1-s)\zeta_F(1-s)$. Denote

$$\zeta_F^*(1) = \mathrm{Res}_{s=1}\, \zeta_F(s) = \pm \frac{2^{r_1}(2\pi)^{r_2} h_F r_F}{w_F \sqrt{|d_F|}}$$

$F = \mathbb{Q}\big(\sqrt{-d}\big)$, $d > 0$, $r_F = 1$, $\zeta_F^*(1) = 2\pi h_F$

$F = \mathbb{Q}\big(\sqrt{d}\big)$, $d > 0$, $\mathcal{O}_F^\times = \pm \varepsilon_F^{\mathbb{Z}}$, ε_F is the fundamental unit, $\zeta_F^*(1) = \frac{h_F \log(\varepsilon_F)}{\sqrt{|d_F|}}$. h_F is a global invariant

$\frac{C}{\mathbb{F}_q}$ is a smooth projective curve, geometrically connected

$$\zeta(C,s) = \prod_{x \in C \text{ closed points}} \frac{1}{1 - N(x)^{-s}} = \prod_x \frac{1}{1 - (q^{-s})^{\deg x}}$$

here $N(x) = [k(x) : k] = q^{\deg s}$. Write $Z(C,t) = \prod_x \frac{1}{1 - t^{\deg x}}$, then $\zeta(C,s) = Z(C, q^{-s})$

$Z(C,t) = P_1 \frac{t}{(1-t)(1-qt)}$, $P_1(t) \in \mathbb{Z}[t]$ of degree $2g$, g is the genus of C

Theorem 1.17 (Weil, Emil Artin, Hasse). $s = 1$, $t = \frac{1}{q}$, $(1-qt)Z(C,t)|_{t=\frac{1}{q}} = P_1 \frac{\frac{1}{q}}{1-\frac{1}{q}} = \frac{h_C}{q-1} q^{1-g}$, here h_C is the class number which is $\#J(C)\big(\mathbb{F}_q\big)$($J$ is the Jacobian), $q-1$ is like w_C, q^{g-1} is like $\sqrt{|d_C|}$

$$\begin{array}{ccc} C & \to & \operatorname{Spec} \mathcal{O}_F \\ \downarrow & & \downarrow \\ \operatorname{Spec} \mathbb{F}_p & \to & \operatorname{Spec} \mathbb{Z} \end{array}$$

We generalize in dimension 2 $\mathcal{E} \subseteq \mathbb{P}^2_{\mathbb{Z}}$

$$\begin{array}{ccc} E & \to & \mathcal{E} \\ \downarrow & & \downarrow \\ \operatorname{Spec} \mathbb{Q} & \to & \operatorname{Spec} \mathbb{Z} \end{array}$$

In Tate's thesis, he dealt with F number field, $\mathbb{F}_q(C)$

$\mathbb{Z} = \prod_p \mathbb{Z}_p$, $A_{\mathbb{Q}}^f = \hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q} \subseteq \prod \mathbb{Q}_p$, $A_{\mathbb{Q}}^f = \big(\hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}\big) \times \mathbb{R} \subseteq \big(\prod \mathbb{Q}_p\big) \times \mathbb{R}$, $\mathbb{A}_F = \mathbb{A}_{\mathbb{Q}} \otimes_{\mathbb{Q}} F$ for a number field F, $\mathbb{G}_m(\mathbb{A}_F) = \mathbb{I}_F$, we have diagonal maps $F \overset{\Delta}{\to} \mathbb{A}_F$, $F^\times \overset{\Delta}{\to} \mathbb{I}_F$, $|\mathbb{G}_m \frac{(\mathbb{A}_F)^1}{\mathbb{G}_m}(F)|$ is finite and arithmetically interesting, here $\mathbb{I}_F^1$ is the kernel of $\mathbb{I}_F \to \mathbb{R}^\times$, $x \mapsto N(x)$

André Weil considered $|\frac{G(\mathbb{A}_F)^1}{G(F)}|$ for $\frac{G}{F}$ is an algebraic group(e.g. $G = \mathrm{SL}_n, \mathrm{SO}_n$), and reintepreted the result of Siegel on quadratic forms

$\frac{E}{\mathbb{Q}}$ is an elliptic curve, since $E(\mathbb{Q}_p) = E(\mathbb{Z}_p)$, $\mathbb{P}^n(\mathbb{Q}) = \mathbb{P}^n(\mathbb{Z})$, $\frac{E(\mathbb{A}_\mathbb{Q})}{E(\mathbb{Q})} = \frac{E(\hat{\mathbb{Z}}) \times E(\mathbb{R})}{E(\mathbb{Z})}$ is not Hausdorff if $\operatorname{rank} E\mathbb{Q} > 0$

BSD conjecture: $\frac{E}{F}$ is an elliptic curve(Abelian variety), $L(E, s) = \prod L_p(E, s)$, $\operatorname{Re} s > \frac{3}{2}$ has a analytic continuation to all of $\mathbb{C}$, $\operatorname{ord}_{s=0} L(F, s) = \operatorname{rank} \mathcal{O}_F^\times$

Weak: $\operatorname{ord}_{s=1} L(E, s) = \operatorname{rank} E(F)$

Strong: $L^\times(E, s)$

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